AISHWARYA NAYAK

Senior AI Engineer | Applied AI | Backend Systems
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SUMMARY

Senior Software Engineer and Applied AI Engineer with 5 years of experience delivering distributed systems, cloud-native services, and AI-driven applications across USA and India teams. Backed by an MS in Data Science from the University at Buffalo (SUNY), with expertise spanning backend engineering, applied AI, frontier-model evaluation, and software-engineering benchmarks. Experienced in building scalable APIs, reproducible testing frameworks, and reliable AI systems while combining strong software engineering fundamentals with modern machine learning practices.

TOOLS & TECNOLOGIES

Programming Languages: Python, Java, Scala, JavaScript, TypeScript, SQL, Bash
Artificial Intelligence & Machine Learning: Large Language Models (LLMs), Agentic AI, AI Evaluation, AI Benchmarking, Prompt Engineering, Multimodal AI, Retrieval-Augmented Generation (RAG), AI Safety, Human-in-the-Loop (HITL), Scientific Integrity Evaluation, Machine Learning, Model Evaluation, Error Analysis
Frameworks & Libraries: LangChain, LangGraph, FastAPI, Pydantic, OpenAI API, Gemini API, Vertex AI
Software Engineering: Backend Development, REST APIs, Microservices, Distributed Systems, Event-Driven Architecture, Software Architecture, Open-Source Development, Feature Development, Bug Fixing, Runtime Debugging
Testing & Benchmark Engineering: Software Engineering Benchmarks, Autonomous Coding Agent Evaluation, Behavioral Testing, Regression Testing, Fail-to-Pass (F2P) Testing, Pass-to-Pass (P2P) Testing, Pytest, Deterministic Evaluation Pipelines, Oracle Implementation Development
Cloud & DevOps: Docker, Kubernetes, Git, GitHub, GitHub Actions, GitHub CLI, Jenkins, AWS, Google Cloud Platform (GCP), Linux
Databases & Search: MySQL, Apache Solr, ChromaDB, Sentence Transformers
Developer Tools: Harbor, n8n, MCP, Slack, JSON, TOML, LaTeX, KaTeX

PROFESSIONAL EXPERIENCE

Handshake AI — AI Evaluation Engineer
Remote (USA & India) | September 2025 – Present

Worked across multiple AI research initiatives focused on frontier large language models, multimodal systems, AI safety, software engineering benchmarks, and autonomous coding agent evaluation.

Software Engineering Benchmark Development

- Designed and authored realistic software engineering benchmark tasks using production-scale open-source repositories to evaluate autonomous AI coding agents.
- Identified real-world bug fixes, feature enhancements, API migrations, and repository-specific engineering challenges suitable for benchmark creation.
- Developed implementation-agnostic benchmark specifications with clearly defined behavioral requirements and evaluation criteria.
- Built deterministic Fail-to-Pass (F2P) and Pass-to-Pass (P2P) test suites supporting multiple valid implementations while preventing solution leakage.
- Implemented production-quality reference solutions, benchmark metadata, debugging hints, evaluation artifacts, and reproducible task packaging.
- Engineered reproducible Docker environments and Harbor-based evaluation pipelines to ensure deterministic execution across diverse repositories.
- Validated benchmark quality against frontier coding models by analyzing agent failure modes, identifying shortcut solutions, and iteratively improving benchmark robustness.

AI Evaluation & Code Quality Assessment

- Evaluated AI-generated software across Python, JavaScript, TypeScript, R, C++, and additional programming languages using execution-based evaluation frameworks.
- Executed generated applications within isolated Docker environments to validate runtime behavior, algorithmic correctness, software design, explanation quality, and instruction adherence.
- Identified runtime failures, logical inconsistencies, hallucinations, incomplete implementations, hidden execution errors, and behavioral regressions.
- Produced reproducible evaluation evidence using execution logs, terminal outputs, structured rubrics, and mirrored evaluation platforms while adapting to evolving project standards.

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Prompt Engineering & Reasoning Evaluation

- Designed challenging multimodal and text-only reasoning tasks to evaluate frontier large language models across computer science and related technical domains.
- Authored image-dependent and text-based prompts following structured evaluation methodologies and standardized quality frameworks.
- Produced complete human-authored reference solutions, structured answer schemas, reproducibility metadata, and detailed technical explanations.
- Validated prompts across multiple frontier models to identify genuine reasoning failures while eliminating ambiguity and retrieval-based errors.
- Reviewed prompts using structured quality rubrics to identify plagiarism, ambiguity, policy violations, invalid evaluation methodology, and overall benchmark quality.

AI Safety & Scientific Integrity Evaluation

- Audited autonomous AI agent trajectories by evaluating problem formulation, planning, implementation, experimentation, and overall methodological integrity.
- Detected scientific integrity violations including data leakage, metric hacking, evaluation contamination, plan-code divergence, constraint violations, tool misuse, and evidence tampering.
- Compared natural-language reasoning with executed code to verify behavioral consistency throughout multi-step experimental workflows.
- Produced evidence-backed integrity assessments and realistic adversarial examples to strengthen AI safety evaluation datasets.

Machine Learning Evaluation

- Evaluated competing machine learning implementation strategies across feasibility, implementation complexity, novelty, and expected impact.
- Analyzed objectives, datasets, evaluation metrics, and implementation approaches to determine technically appropriate algorithmic solutions.
- Identified invalid evaluation tasks involving duplicate algorithms, non-machine-learning_objectives, and policy violations.

Multimodal AI & Data Curation

- Contributed to multimodal dataset generation initiatives through standardized collection, quality assurance, and compliance workflows.
- Captured high-quality multimodal datasets following strict privacy, safety, consent, and geographic requirements.
- Performed quality validation to maximize downstream usability for computer vision, navigation, and multimodal AI model training.

Reality AI Lab — Generative AI Engineer
Buffalo, New York, USA | Mar 2025 – Sep 2025

- Developed a retrieval-augmented AI teaching assistant enabling interactive academic dataset querying using Python, LangChain, and OpenAI APIs.
- Built scalable ingestion and preprocessing pipelines supporting long-context LLM applications, reducing input errors by 15%.
- Deployed models using GCP Vertex AI and containerized services, automating model training and production inference workflows.
- Integrated logging, evaluation, and monitoring frameworks to improve reliability and accelerate root-cause analysis.
- Collaborated with backend engineers to design modular APIs supporting knowledge graph updates and prompt orchestration workflows.
- Contributed to the end-to-end lifecycle of AI applications spanning data ingestion, model deployment, inference, and observability.

Mocapay — Software Engineer, Backend
Delhi, India | Aug 2021 – Jun 2023

- Developed and maintained reactive Scala microservices supporting fintech products generating over $10M in annual revenue.
- Designed and implemented RESTful APIs for large-scale transaction processing systems, improving reliability and customer experience.

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- Integrated Apache Kafka for asynchronous communication and event-driven workflows, enhancing real-time transaction capabilities and system resilience.
- Streamlined CI/CD pipelines using Jenkins and Docker, reducing deployment time by 30%.
- Enhanced platform robustness and security in accordance with Vanta recommendations, contributing to SOC 2 compliance.
- Improved search capabilities by integrating Apache Solr, increasing search efficiency by 25%.
- Led code reviews and promoted software engineering best practices to improve maintainability and code quality.

Tata Consultancy Services (TCS) — Software Engineer, Backend
Indore, India | Jan 2020 – Jul 2021

- Led development of a finance platform supporting management of over 1,000 contracts and improving operational efficiency.
- Designed backend services and RESTful APIs for contract creation, validation, and payment calculations.
- Managed cloud deployments on AWS and GCP, reducing infrastructure costs by 15%.
- Optimized SQL queries and improved database performance by 20%.
- Implemented security controls and role-based access mechanisms using Fortify and SonarQube, reducing vulnerabilities by 40%.
- Automated testing and deployment pipelines using Jenkins, significantly shortening release cycles.
- Collaborated with cross-functional teams during sprint planning, backlog refinement, and product roadmap discussions.

SELECTED PROJECTS

Enterprise Knowledge Copilot | RAG Knowledge Assistant [GitHub]

Tech Stack: Python, LangChain, Vector Databases, OpenAI APIs, FastAPI

- Built an enterprise-grade Retrieval-Augmented Generation (RAG) platform for contextual question answering and knowledge discovery across organizational documents.
- Designed scalable ingestion and retrieval pipelines supporting structured and unstructured data sources.
- Implemented semantic search and source-grounded response generation to improve answer quality and reduce hallucinations.
- Developed modular APIs and extensible workflows supporting future tool integrations and agent-based interactions.
- Enabled efficient knowledge access and productivity improvements through context-aware conversational interfaces.

CodeSage | Multi-Agent Repository Intelligence Platform [GitHub]

Tech Stack: Python, LangGraph, Pydantic, GitHub APIs

- Developed a LangGraph-based multi-agent system for automated repository analysis and architecture understanding.
- Built specialized agents for dependency inspection, quality assessment, security analysis, and documentation generation.
- Implemented shared-state orchestration and structured outputs using Pydantic to improve reliability and maintainability.
- Automated report generation workflows to accelerate onboarding and analysis of unfamiliar codebases.
- Designed extensible agent workflows for evaluating repositories and enhancing developer productivity.

Transcript Intelligence | AI-Powered Transcript Analytics Platform [GitHub]

Tech Stack: Python, Jupyter Notebook, Pandas, OpenRouter, Claude Sonnet 4.5, structured JSON outputs, Matplotlib

- Built an AI-powered transcript analytics pipeline that classifies business conversations, analyzes sentiment, and extracts cross-functional insights from unstructured call data.
- Designed a hybrid topic classification workflow combining LLM-generated taxonomy discovery with human review to improve consistency and business relevance.
- Implemented multi-dimensional sentiment analysis covering customer sentiment, urgency, resolution status, and sentiment trajectory rather than relying on a single polarity label.
- Developed a cross-functional incident timeline that linked support, external customer, and internal engineering conversations to reconstruct outage impact across teams.
- Added reproducible outputs, validation artifacts, and notebook-based reasoning to support transparent analysis and portfolio presentation.

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FlowForge AI | AI Workflow Automation Platform [GitHub]

Tech Stack: n8n, Python, APIs, LLM Integrations

- Developed an AI-driven workflow automation platform enabling orchestration of multi-step tasks through modular and event-driven pipelines.
- Built reusable workflows integrating external APIs and large language models to automate repetitive processes and improve productivity.
- Designed extensible automation patterns supporting tool interactions, conditional execution, and multi-stage task coordination.
- Leveraged low-code orchestration principles to simplify integrations and accelerate workflow development.
- Enhanced operational efficiency through reusable components and scalable workflow design.

Log-Report | AI-Powered Log Analysis and Incident Reporting System [GitHub]

Tech Stack: Python, OpenAI API, LangChain, Pydantic, JSON, Logging, Regex, Async Processing

- Built an AI-powered log analysis pipeline that transforms raw logs into structured incident reports.
- Implemented intelligent parsing, error clustering, root-cause extraction, severity classification, and timeline construction.
- Generated Markdown and JSON reports for production incident review, debugging, and support automation.

EDUCATION

Master of Science (MS) in Data Science — Aug 2023 - Feb 2025

University at Buffalo (State University of New York, SUNY Buffalo) — New York, US

Bachelor of Technology (B.Tech) in Electronics and Telecom Engineering — Jul 2015 - Apr 2019

Symbiosis International University — Pune, IN